

Project Initialization and Planning Phase

Date	15 July 2024
Team ID	*****
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine-learning-based credit model and real-time decision-making.

Project Overview

Field	Details
Objective	Revolutionize the loan approval process by implementing a machine-learning model that assesses applicant credibility faster and more consistently than manual review alone.
Scope	Comprehensively assess and enhance the loan approval process: intake form, feature encoding, trained RandomForest model, and an instant approve/decline result page, with room to add manual-review escalation later.

Problem Statement

Field	Details
Description	Manual loan approval is time-consuming and prone to inconsistency, which hurts operational efficiency and customer satisfaction.
Impact	Solving this improves operational efficiency, reduces risk from inconsistent manual calls, and improves customer satisfaction and organizational outcomes.

Proposed Solution

Field	Details
Approach	Employ machine learning to analyze and predict creditworthiness from applicant income, credit history, employment, and household details, creating a dynamic, repeatable loan-approval read.
Key Features	• Machine-learning-based credit assessment model (RandomForestClassifier) • Real-time decision-making through a simple web form • Designed to be retrained as new applications and outcomes accumulate

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware – Computing Resources	CPU/GPU specifications, number of cores	Standard CPU (no GPU required for RandomForest training/inference)
Hardware – Memory	RAM specifications	8 GB
Hardware – Storage	Disk space for data, models, and logs	1 GB (dataset + pickled model are both small)
Software – Frameworks	Python frameworks	Flask
Software – Libraries	Additional libraries	scikit-learn, pandas, numpy
Software – Development Environment	IDE	Jupyter Notebook, VS Code / PyCharm
Data – Source, size, format	Dataset details	Kaggle “Loan Prediction” dataset, 614 records, CSV